THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS UNIT TEST SET A

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. $\sqrt{1089}$ is equal to: 1

(a) 32 **(b)** 34 **(c)** 33 **(d)** 35

2. Which of the following is a perfect square? 1

(a) 300 **(b)** 250 **(c)** 225 **(d)** 230

3. The value of 17^2 is: 1

(a) 272 **(b)** 306 **(c)** 34 **(d)** 289

4. $3^4 \times 3^6$ equals: 1

(a) 3^{10} **(b)** 9^{10} **(c)** 3^{24} **(d)** 3^{-2}

5. The value of $\left(\frac{1}{3}\right)^{-2}$ is: 1

(a) 6 **(b)** 9 **(c)** $\frac{1}{9}$ **(d)** $-\frac{1}{9}$

6. 2^{10} is closest to: 1

(a) 1000 **(b)** 2000 **(c)** 10000 **(d)** 100

7. The Roman numeral XCIV stands for: 1

(a) 104 **(b)** 84 **(c)** 94 **(d)** 95

8. In Roman numerals, a smaller symbol written before a larger one is: 1

(a) multiplied **(b)** ignored **(c)** subtracted **(d)** added

9. Three angles of a quadrilateral are 80°, 100° and 90°. The fourth angle is: 1

(a) 100° **(b)** 80° **(c)** 110° **(d)** 90°

10. Every square is a: 1

(a) kite only **(b)** scalene triangle **(c)** rhombus **(d)** trapezium only

11. A number ending in 00 is always divisible by: 1

(a) 11 **(b)** 3 **(c)** 7 **(d)** 100

12. The Kaprekar constant for four-digit numbers is: 1

(a) 6174 **(b)** 1089 **(c)** 1729 **(d)** 1234

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. Is 1225 a perfect square? Give its square root. 2

MASTERMIND

14. Evaluate $(-1)^{101}$ and $(-1)^{100}$. 2

15. Why is the Hindu number system considered a great invention? 2

16. Why is every square a rectangle? 2

17. State the divisibility test for 3. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Find $\sqrt{2025}$ using prime factorisation. 3

19. Write 3888 in Roman numerals and explain the grouping. 3

20. Prove that the opposite angles of a parallelogram are equal. 3

21. Find the smallest number divisible by 6, 9 and 15. 3

22. Diagonals of a rhombus are 16 cm and 30 cm. Find the side and perimeter. 3

OR

In parallelogram ABCD, $\angle A = (3x + 10)^\circ$ and $\angle B = (2x - 10)^\circ$. Find all angles. 3

23. Find all digits that can replace the blank in 9_36 to make it divisible by 3. 3

OR

Find the LCM of 24, 36 and 40 and the HCF of 24 and 36. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) Find the value of $\left(\frac{2}{3}\right)^{-3} \times \left(\frac{2}{3}\right)^{2}$. 5

(ii) Simplify $\dfrac{(25)^3 \times 5^2}{5^{10}}$.

OR

(i) The Milky Way is about 10^5 light-years wide and one light-year is about 9.46×10^{12} km. Estimate its width in km. 5

(ii) Express in standard form.

25. **Case study:** Three traffic lights change every 40 s, 60 s and 90 s respectively and all turn green together at 9:00 a.m. 5

(i) After how many seconds will they next change together?

(ii) At what time?

(iii) How many times will they change together between 9:00 and 10:00, counting both 9:00 and 10:00?

OR

(i) State the divisibility tests for 2, 4, 5 and 8. 5

(ii) Test 3,15,000 and 59,32,816 for these.

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND